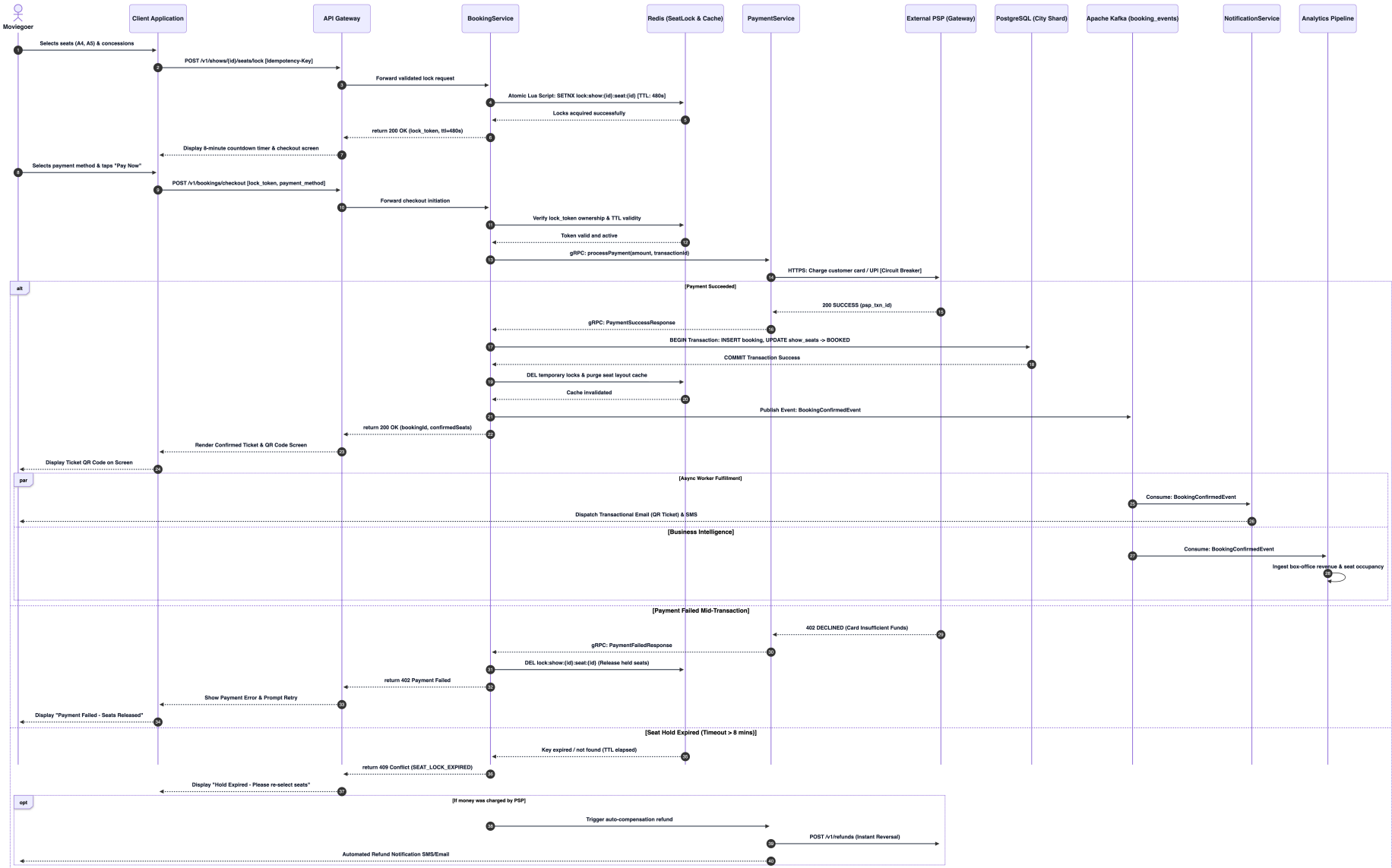


End-to-End System Sequence: Full Booking Lifecycle & Failure Paths

Trace: User → UI → API Gateway → BookingService → Redis SeatLock → PaymentService → PostgreSQL (City Shard) → Kafka (booking_events) → NotificationService / Analytics.

Failure Recovery: Explicit alternate branches for Payment Declined and Seat Lock Expiration with automatic compensation refunds.



Lifecycle Execution Phases & Automated Failure Recovery Matrix

Detailed matrix mapping normal execution phases and automated compensation logic for critical system failures.

Phase / Failure Scenario	Trigger & Detection	System Action & Invariant Preservation	Compensation / User Outcome
Phase 1: Seat Locking	User selects seats on UI map; submits lock request.	Redis executes atomic SETNX with 480s TTL; marks seat HELD.	Success returns lock_token; conflict returns 409 if held by another.
Phase 2: Payment Settle	User authorizes payment via card, UPI, or wallet.	PaymentService executes charge via Circuit Breaker to external PSP.	Returns gateway transaction ID; idempotency key prevents double charge.
Phase 3: Persistence	Successful payment response received from gateway.	Atomic PostgreSQL write on city shard: updates show_seats to BOOKED.	Redis temporary hold keys deleted; seat layout cache invalidated.
Phase 4: Async Delivery	PostgreSQL transaction commits successfully.	BookingService emits BookingConfirmedEvent to Kafka event bus.	NotificationService dispatches QR ticket via Email/SMS; Analytics updates.
Failure: Payment Decline	PSP returns 402 Declined or gateway timeout.	BookingService catches failure; skips PostgreSQL transaction commit.	Redis hold key deleted immediately (or retry permitted within TTL).
Failure: Hold Expiration	Checkout duration exceeds 8 minutes (Redis TTL elapsed).	BookingService validates lock token before charge; detects missing key.	Rejects transaction (409 Conflict); auto-refunds customer if debited.
Failure: Consumer Crash	Notification worker pod OOMs or node fails.	Kafka detects missed heartbeat; triggers partition group rebalance.	Healthy worker resumes from last committed offset; zero messages lost.
Failure: Gateway Outage	Downstream email/SMS provider returns 5xx errors.	Worker retries with exponential backoff; moves to DLQ after 3 attempts.	Core booking remains confirmed; DLQ replay worker redelivers on recovery.

DevOps Deployment & Rollback Blueprint:

- **Containerized Stack:** Microservices run on Kubernetes (EKS) with HPA autoscaling based on CPU (>70%) and RPS.
- **Blue-Green Strategy:** New releases deployed to idle Green environment; canary traffic routed (10% → 100%) through Application Load Balancer.
- **Instant Rollback:** If error rates exceed 0.5% or p99 latency exceeds 2.0s, traffic reverts to Blue in < 5 seconds with zero customer impact.